48730-32548, Cyber Security Week-5
Understanding TCP/IP based Attacks

Task 1: SYN Flooding Attack

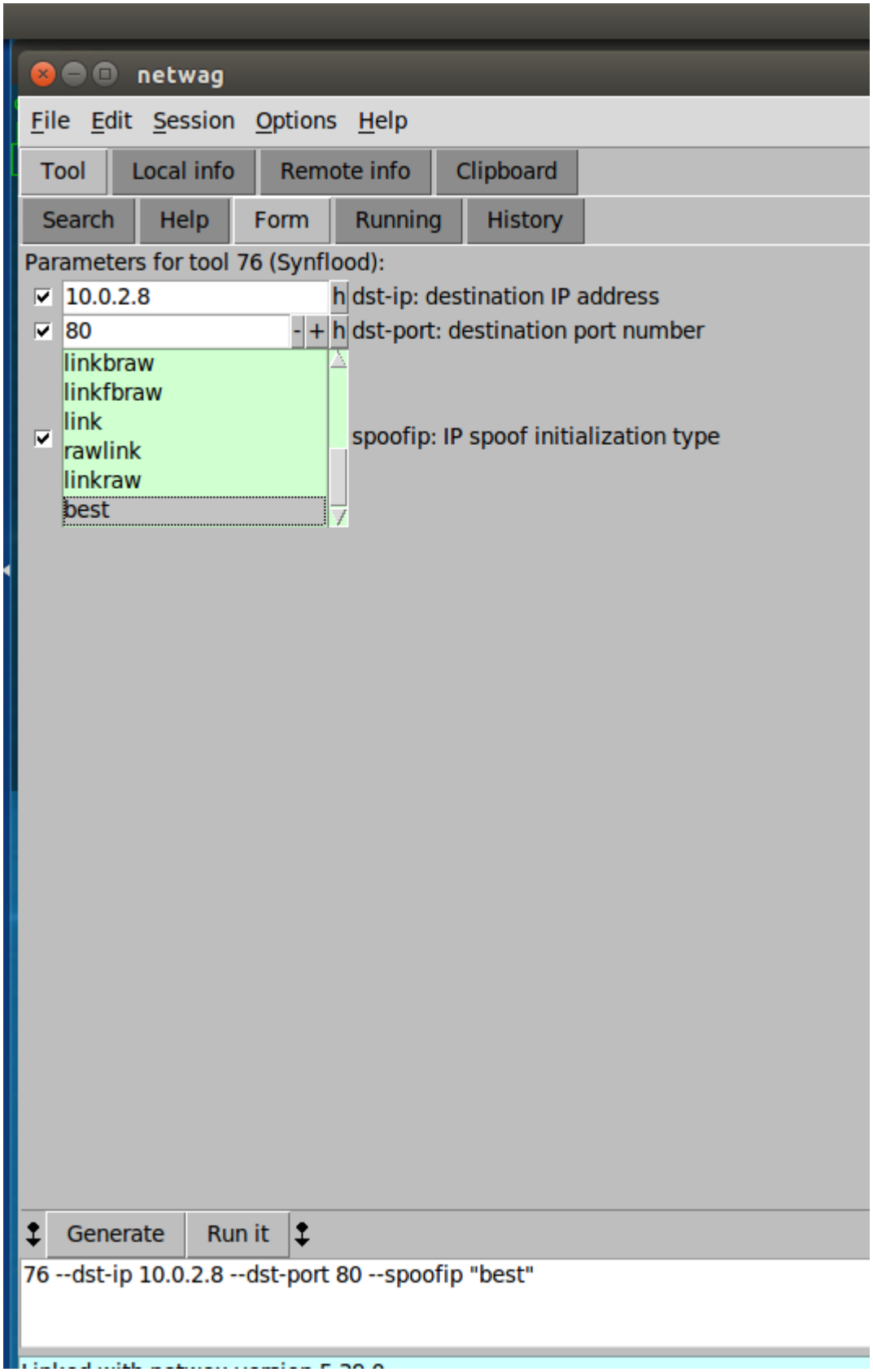
Steps:

1. As SYN flood produces a lot of traffic in the VM due to high speed, Wireshark might
crash. It is recommended to use “tshark” by entering “sudo tshark” on Terminal of the Client VM.

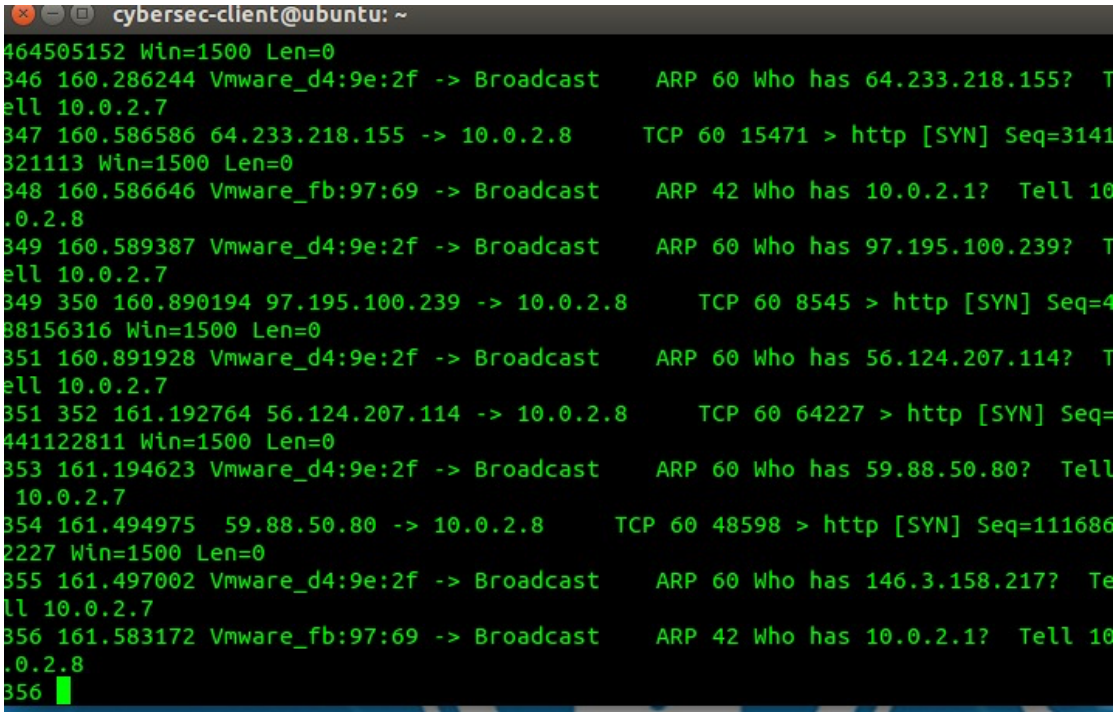
2. Then go to Attacker VM and start Netwag by entering “sudo netwag” on the Terminal.

3. Select **76: Synflood.**

4. Enter the details and click run it. **(Screenshot required)**


netwag
File Edit Session Options Help
Tool
Local info
Remote info
Clipboard
Search
Help
Form
Running
History
Parameters for tool 76 (Synflood):
10.0.2.8
h
dst-ip: destination IP address
80
- + h
dst-port: destination port number
linkbraw
linkfbraw
link
rawlink
linkraw
best
spoofip: IP spoof initialization type
Generate
Run it
76 --dst-ip 10.0.2.8 --dst-port 80 --spoofip "best"

5. Observe the captured packets on Client VM. **(Screenshot required)**



```
cybersec-client@ubuntu: ~
464505152 Win=1500 Len=0
346 160.286244 Vmware_d4:9e:2f -> Broadcast    ARP 60 Who has 64.233.218.155?  T
ell 10.0.2.7
347 160.586586 64.233.218.155 -> 10.0.2.8     TCP 60 15471 > http [SYN] Seq=3141
321113 Win=1500 Len=0
348 160.586646 Vmware_fb:97:69 -> Broadcast    ARP 42 Who has 10.0.2.1?  Tell 10
.0.2.8
349 160.589387 Vmware_d4:9e:2f -> Broadcast    ARP 60 Who has 97.195.100.239?  T
ell 10.0.2.7
349 350 160.890194 97.195.100.239 -> 10.0.2.8     TCP 60 8545 > http [SYN] Seq=4
88156316 Win=1500 Len=0
351 160.891928 Vmware_d4:9e:2f -> Broadcast    ARP 60 Who has 56.124.207.114?  T
ell 10.0.2.7
351 352 161.192764 56.124.207.114 -> 10.0.2.8     TCP 60 64227 > http [SYN] Seq=
441122811 Win=1500 Len=0
353 161.194623 Vmware_d4:9e:2f -> Broadcast    ARP 60 Who has 59.88.50.80?  Tell
 10.0.2.7
354 161.494975  59.88.50.80 -> 10.0.2.8     TCP 60 48598 > http [SYN] Seq=111686
2227 Win=1500 Len=0
355 161.497002 Vmware_d4:9e:2f -> Broadcast    ARP 60 Who has 146.3.158.217?  Te
ll 10.0.2.7
356 161.583172 Vmware_fb:97:69 -> Broadcast    ARP 42 Who has 10.0.2.1?  Tell 10
.0.2.8
356
```

Questions:

1. Observe the attack and take screenshots of the attack scenario

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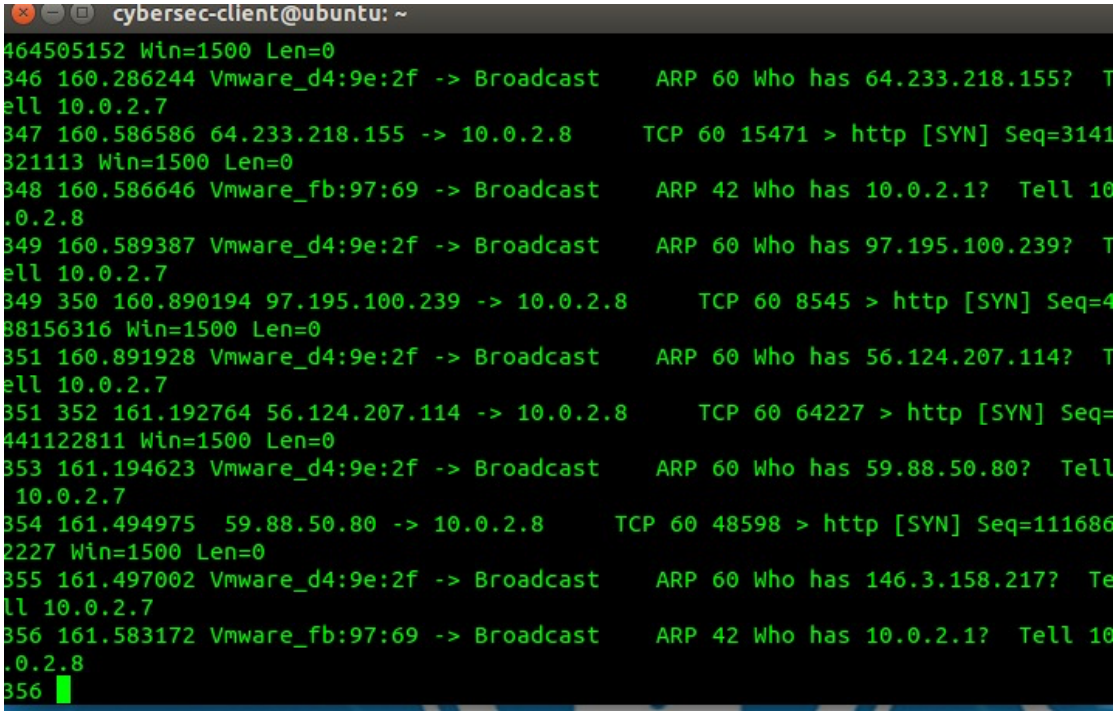


2. Comment on your observation.

The attacker kept spamming SYNs. This demonstrates that there is an attacker who is flooding the victim with excessive requests. It will stop real clients from connecting.

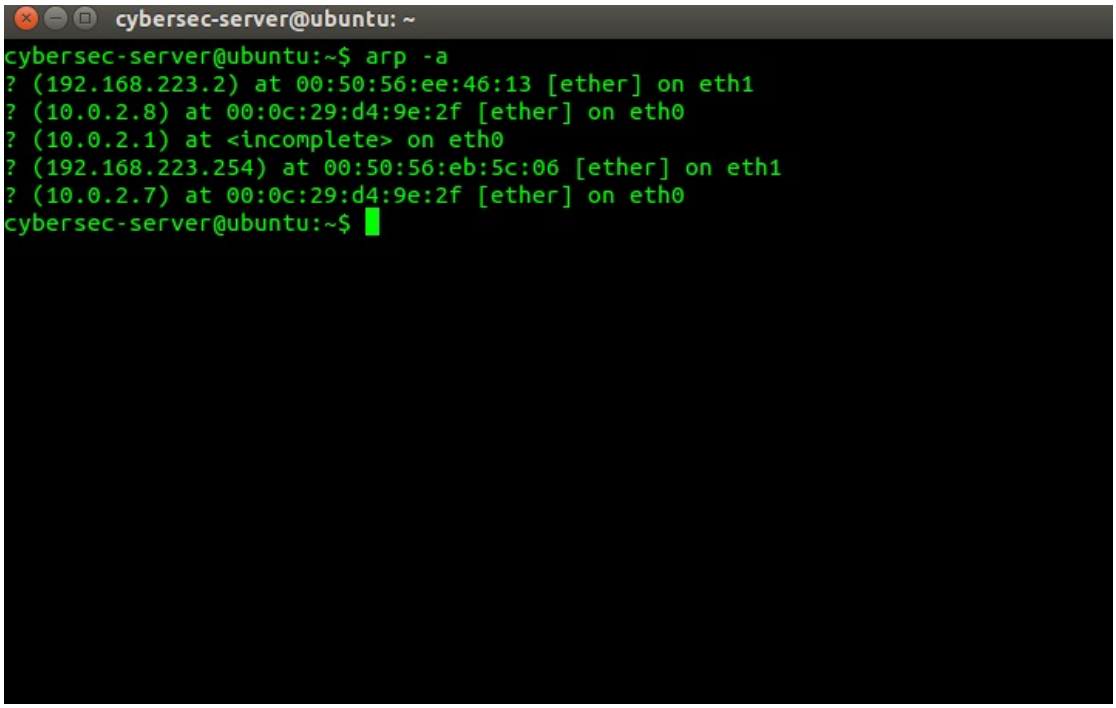
3. Categorize this attack in terms of severity and how it is linked to the DoS attack

This is a Dos attack. It makes servers busy with fake requests and blpocking users from connecting.

Task 2: ARP cache poisoning

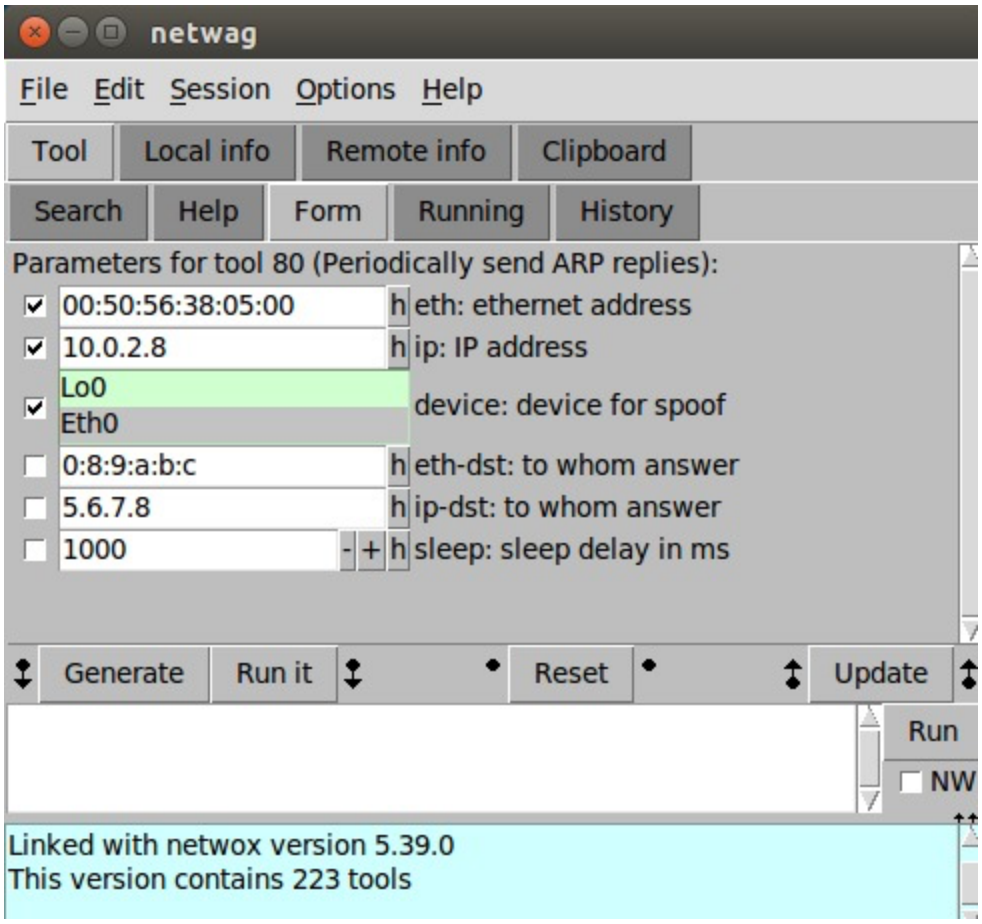
Steps:

1. In Server VM, run “arp -a” on terminal to get the ARP Information (MAC Table). **(Screenshot Required)**



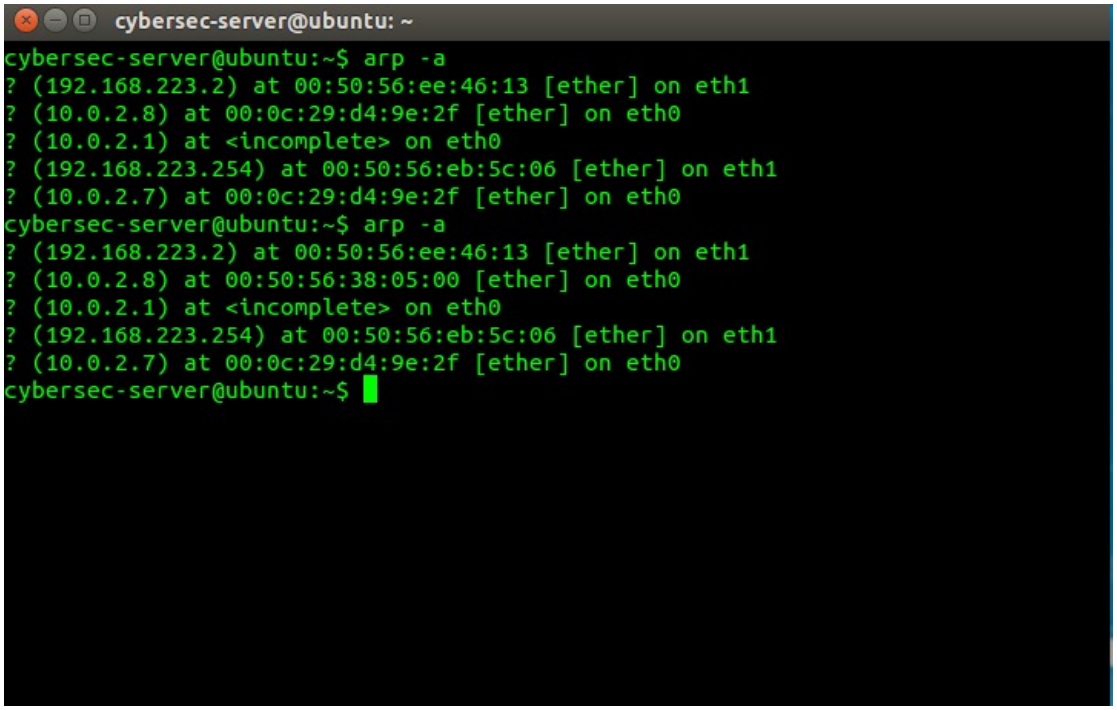
2. Open Netwag on Attacker VM by entering “sudo netwag” on the Terminal.
3. Select **Tool 80: Periodically Send ARP Replies**.

4. Add the fake MAC address and IP address and select the interface. **(Screenshot Required)**



5. Click “Run It”.

6. Check the MAC Table on Server VM and look for the change in MAC address for IP address provided in the Netwag tool 80. **(Screenshot required)**



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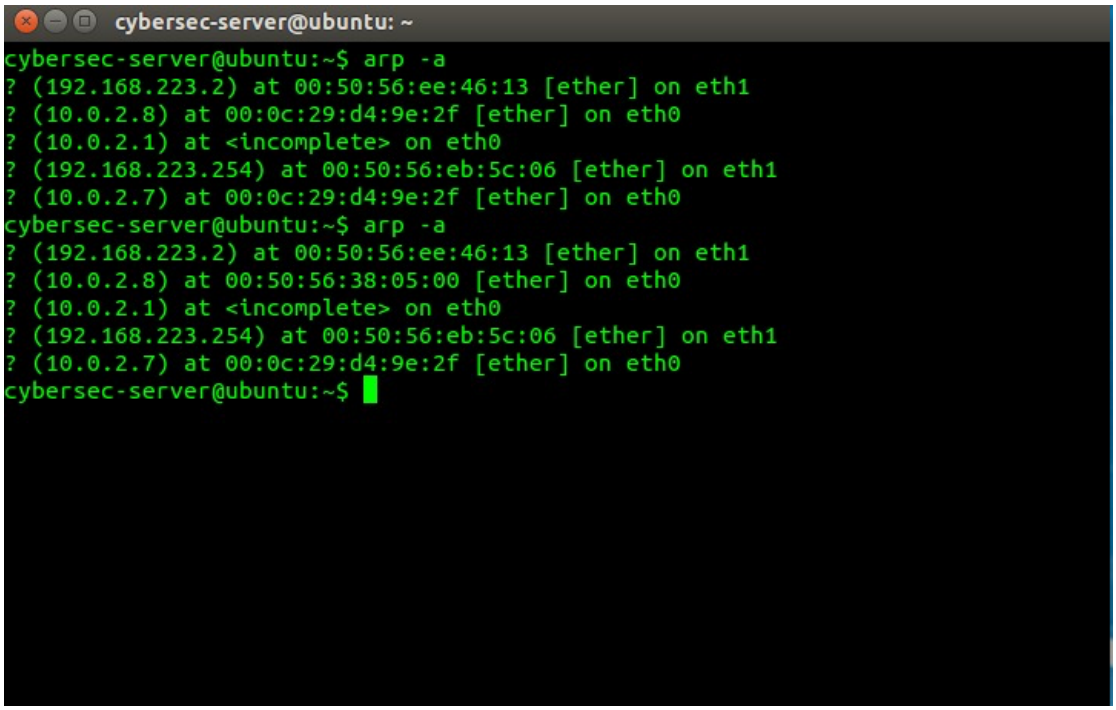


2. Comment on your observation.

The attacker tricked the server into storing a wrong MAC gateway, which could redirect or block traffic

3. Briefly describe how you can mitigate this attack.

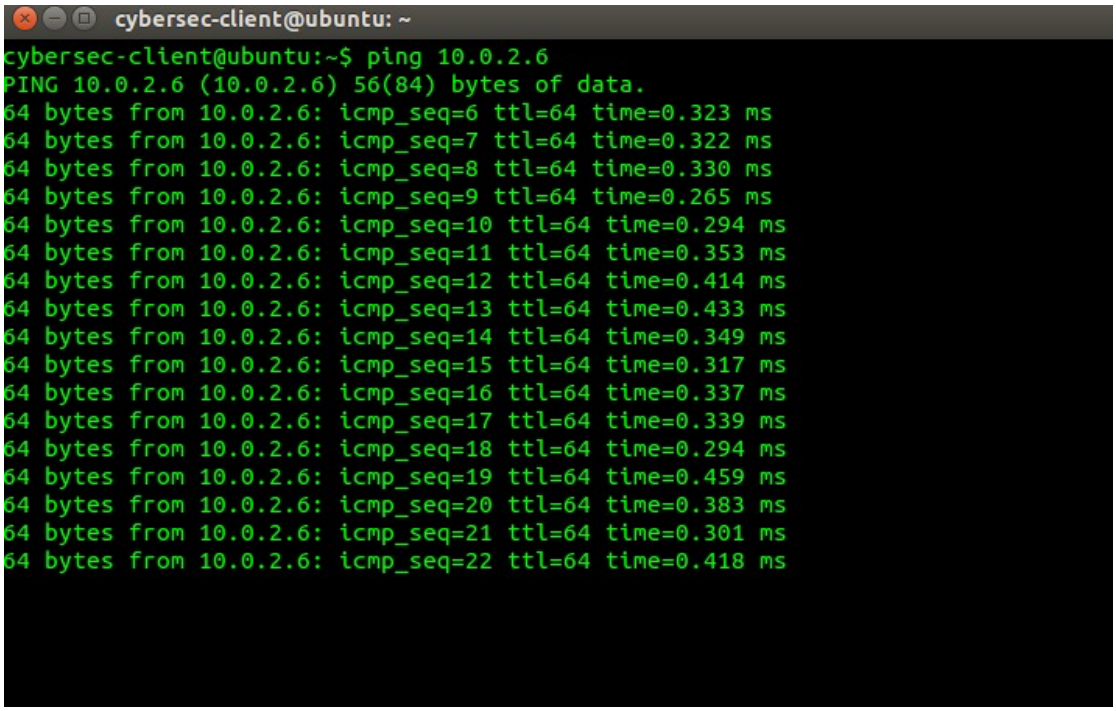
This attack can be mitigated by configuring static ARP entries for critical systems

Task 3: ICMP Redirect Attack

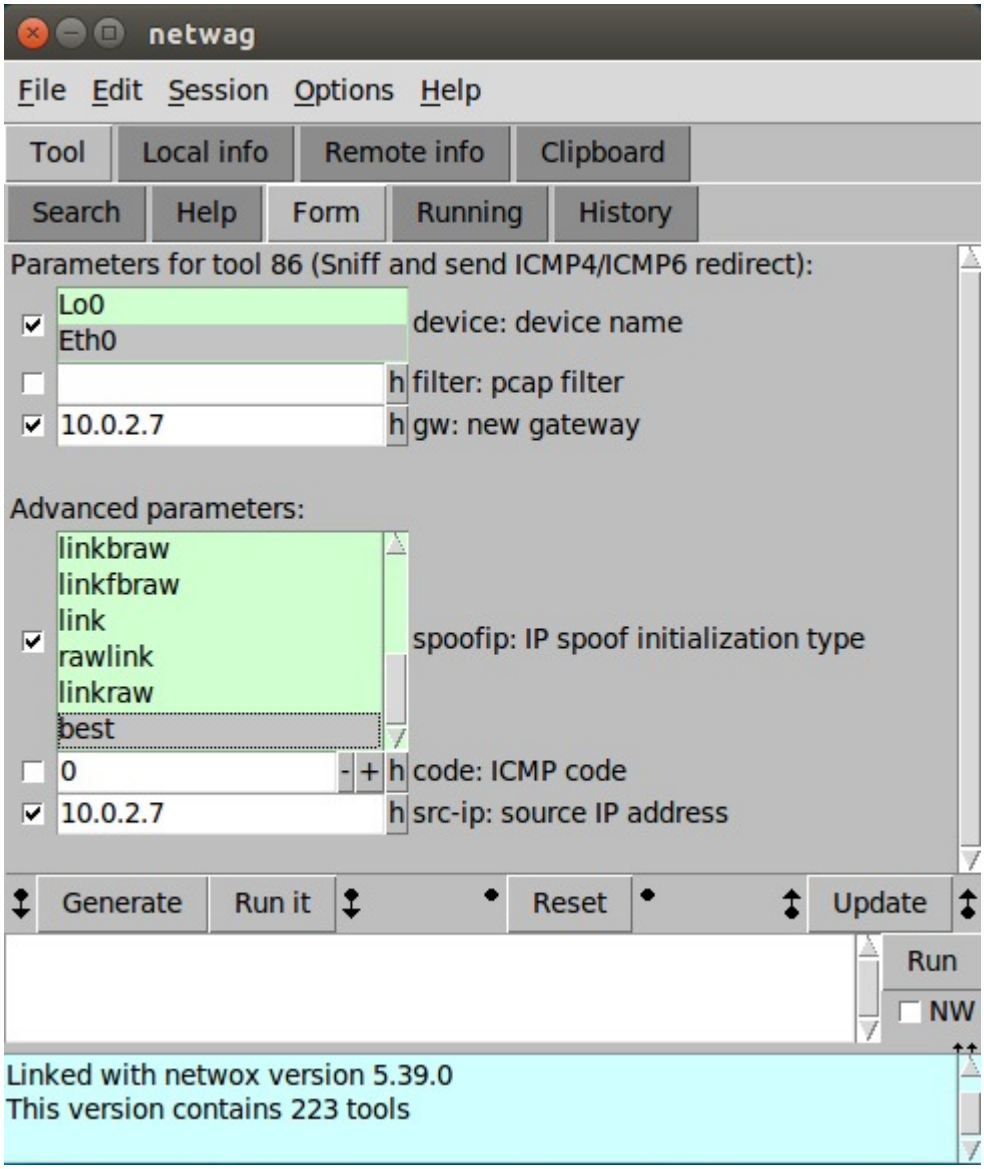
Steps:

1. Open Wireshark on Client VM by entering “sudo wireshark” in the terminal.
2. Select the interface and set “icmp” as filter and click “Apply”.

3. Open a new terminal on Client VM and ping the server. **(Screenshot required)**

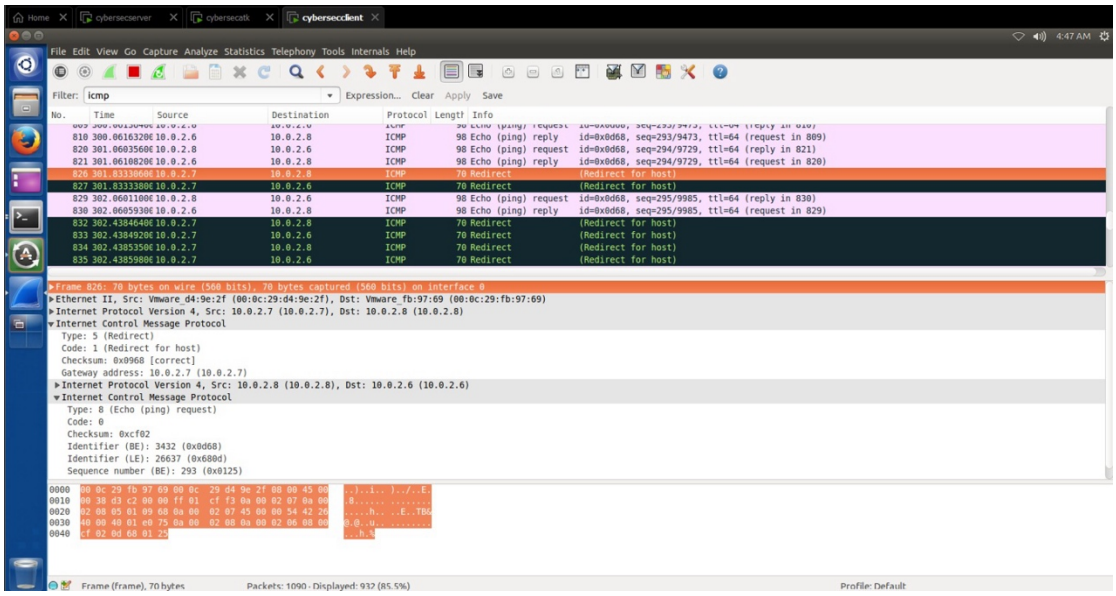


4. Go to Attacker VM, run Netwag using the terminal.
5. Select the interface and “spoofip: IP spoof initialization type”. Input the required IP address into “gw: new gateway” and “src-ip: source IP address”. **(Screenshot Required)**



6. Click “Run It”.

7. Go back to Client VM, check the Wireshark output. **(Screenshot Required)**



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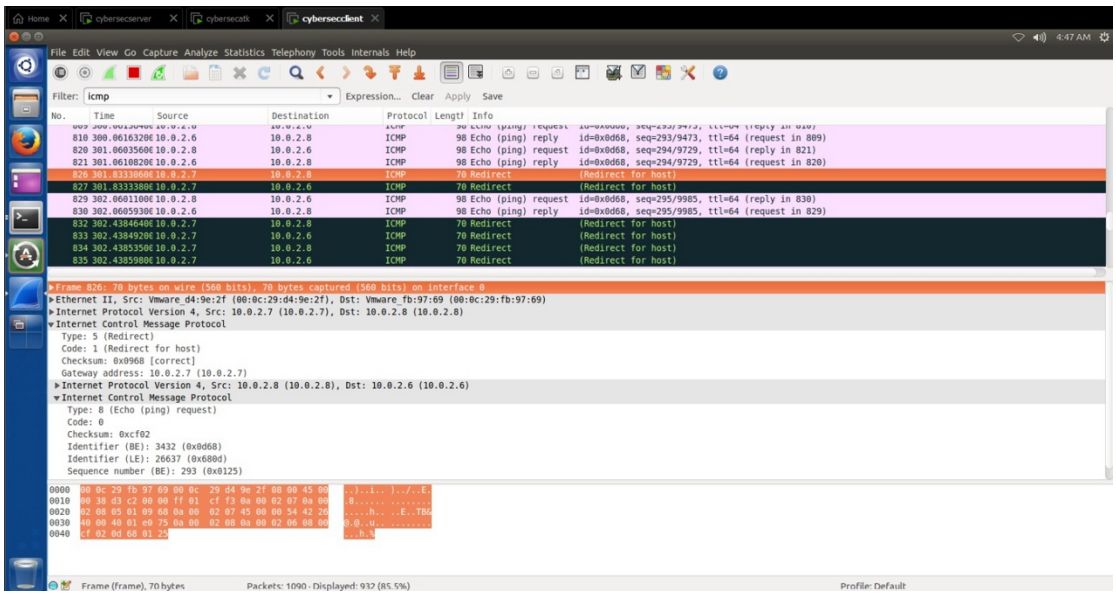

2. Comment on your observation.

The attacker tricked the client into changing its route.

3. Briefly describe how you can mitigate this attack

We can mitigate this kind of attack by using static routing where possible, deploying secure routing protocols with authentication